

Embedded Communication Protocols - Interview Quiz & Answers

1. UART doesn't use a clock line. How do two devices know when to sample data correctly?

[Correct] Correct Answer: C - Baud rate agreement

2. If your UART is set to 9600 baud, how long does it take to transmit one byte (8 bits, no parity, 1 stop)?

[Correct] Correct Answer: D - 1.04 ms

3. What happens if two UART devices have mismatched baud rates (e.g., 9600 and 115200)?

[Correct] Correct Answer: A - Data gets corrupted

4. In I2C, what does it mean when SDA is pulled low during data transfer?

[Incorrect] Your Answer: A - Data is being written

[Correct] Correct Answer: D - Slave ACK/NACK

5. Two I2C devices have same address. What can you do?

[Correct] Correct Answer: C - Use a mux or change one sensor's address

6. Why is I2C not used for long-distance PCB connections?

[Incorrect] Your Answer: B - High current draw

[Correct] Correct Answer: C - Capacitive loading and slow rise times

7. SPI giving garbage data. What's the first thing to check?

[Correct] Correct Answer: A - SPI clock polarity and phase (CPOL/CPHA)

8. Why is SPI full duplex?

[Correct] Correct Answer: B - It sends and receives simultaneously

9. Sensor data valid on rising edge - which SPI mode?

[Correct] Correct Answer: C - CPOL=0, CPHA=1

10. Interface 5 I2C devices with same address?

[Correct] Your Answer: Use I2C mux like TCA9548A

11. Debug system hang with UART/I2C/SPI?

[Correct] Your Answer: Use printf logs, LED blinks, disable peripherals one-by-one

12. When to use SPI vs I2C?

[Correct] Your Answer: SPI for real-time speed, I2C for simple multi-device use